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METHOD FOR PRODUCING A CAST VEHICLE WHEEL

Abstract

A method of manufacturing a cast vehicle wheel including a wheel trim cover attached thereto. A removable first tool insert is installed into a mold assembly. The tool insert includes a mounting feature that communicates with a cavity of the mold assembly to form a mating feature in the wheel. A wheel trim cover is provided and is attached to the wheel at the mating feature via a coupler formed in the wheel trim cover such that the relative position of the coupler and the mating feature positions the wheel trim cover relative to the wheel in a first predetermined position. After casting, the positional dimensions of the mating feature of the wheel relative to the coupler of the wheel trim cover is evaluated to determine the positional accuracy of the mounting feature of the first tool insert. If necessary, secondary inserts with improved mating features are provided and exchanged.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention relates in general to the production or manufacturing of a cast vehicle wheel, and in particular to an improved manufacturing process for producing a cast vehicle wheel having a plurality of wheel trim covers attached thereto.

[0002] Wheels for automotive vehicles may be produced by various methods. For example, the vehicle wheels may be produced within a mold assembly as a generally single cast component made from aluminum or alloys thereof. Some vehicle wheels may include additional trim panels or trim covers attached to the vehicle wheels post casting of the wheels. The trim covers may be provided for aesthetic reasons or for aerodynamic properties. These trim covers may be made of relatively lightweight plastic to reduce the overall weight of the wheel. The trim covers are typically fastened to the wheel such as by threaded fasteners.

[0003] The design and manufacture of the cast wheels may include integrally formed mating features to assist in the alignment, positioning and/or the centering of the trim covers relative to the wheel. The integrally formed mating features are adapted to receive corresponding mating features or couplers of the trim covers. The couplers formed on the trim covers provide an integral structural feature for properly positioning and joining the trim covers to the wheel. For example, the couplers may be in the form of protrusions integrally formed on the rear side of the trim covers which align with and mate with corresponding recesses formed on the front face of the wheel functioning as mating features of the wheel. Once properly positioned on the wheel, the trim covers may be more securely fastened to the wheel, such as by threaded fasteners.

[0004] Due to manufacturing tolerances of the cast wheel and/or the trim covers, the trim covers may not properly center or align properly with the wheel. To correct this issue, the mold assembly for the trim covers and/or the cast wheel may need to undergo additional tooling to correct the mating feature portions of the mold assembly. However, this additional tooling can be quite expensive and time-consuming resulting in delays and additional costs in the production of the wheels.

SUMMARY OF THE INVENTION

[0005] The present invention relates to an improved method for producing a cast vehicle wheel as illustrated and/or described herein.

[0006] According to one embodiment, the method for producing a cast vehicle wheel may comprise, individually and/or in combination, one or more of the following steps, features, elements, and/or advantages: a method of manufacturing a cast wheel including at least one wheel trim cover attached thereto, wherein the method comprises the steps of: (a) providing a wheel mold assembly having a cavity defining the shape of the wheel; (b) providing a removable first tool insert adapted to be installed into the mold assembly, wherein the first tool insert includes a mounting feature that communicates with the cavity of the mold assembly to form a mating feature in the wheel when the wheel is cast; (c) installing the first tool insert into the mold cavity; (d) casting the wheel by introducing a casting material into the cavity of the mold assembly, thereby forming the mating feature in the wheel; (e) removing the wheel from the mold assembly; (f) providing at least one wheel trim cover adapted to be attached to the wheel at the mating feature of the wheel via a coupler formed in the wheel trim cover such that the relative position of the coupler and the mating feature positions the wheel trim cover relative to the wheel in a first predetermined position; (g) attaching the wheel trim cover onto the wheel at the location of the coupler and the mating feature; (h) evaluating the positional dimensions of the mating feature of the wheel relative to the coupler of the wheel trim cover to determine the positional accuracy of the mounting feature of the first tool insert; (i) subsequent to step (h), providing a second tool insert having a second mounting feature dimensionally different from the mounting feature of the first tool insert which provides improved positional accuracy of the wheel trim cover relative to the wheel in a second predetermined position; (j) replacing the first tool insert from the mold cavity and installing the second tool insert into the mold assembly; (k) casting a second wheel by introducing a casting material into the cavity of the mold assembly; and (l) attaching the wheel trim cover onto the second wheel at the location of the coupler and the second mating feature.

[0007] According to this embodiment, the coupler of the wheel trim cover is in the form of a protrusion, and wherein the mating feature of the wheel is a recess receiving the protrusion when the wheel trim cover is attached to the wheel.

[0008] According to this embodiment, in step (d) during the casting process, the formation of the profile of the mating feature of the wheel corresponds to the profile of the mounting feature of the first tool insert.

[0009] According to this embodiment, in step (g) the wheel trim cover is attached onto the wheel by at least one fastener.

[0010] According to this embodiment, the fastener is a threaded bolt.

[0011] According to this embodiment, wherein prior to step (g) a hole is formed in the wheel at the mating feature of the wheel for receiving the threaded bolt.

[0012] According to this embodiment, the hole is formed by a machining or drilling process after the wheel is cast.

[0013] According to this embodiment, the hole is formed in the casting process.

[0014] According to this embodiment, the wheel trim cover is provided with internal threads formed therein for receiving the threaded bolt.

[0015] According to this embodiment, the coupler of the wheel trim cover is provided with the internal threads.

[0016] According to this embodiment, the coupler is a protrusion formed in the wheel trim cover.

[0017] According to this embodiment, the mating feature of the wheel is a recess receiving the protrusion of the wheel trim cover.

[0018] According to this embodiment, the protrusion is cylindrically shaped.

[0019] According to this embodiment, the recess is cylindrically shaped.

[0020] According to this embodiment, in step (l) the wheel trim cover is attached onto the second wheel by at least one fastener.

[0021] According to this embodiment, in step (h), evaluating the positional dimensions of the mating feature of the wheel relative to the coupler of the wheel trim cover includes measuring the position of the mating feature of the wheel relative to the position of the coupler of the wheel trim cover.

[0022] According to this embodiment, the tool insert includes a keyway which mates with a key portion of the mold assembly to prevent rotation of the tool insert within the mold assembly.

[0023] According to this embodiment, the coupler of the wheel trim cover includes a distal surface, and wherein the distal surface contacts the wheel when the wheel trim cover is attached to the wheel.

[0024] According to this embodiment, the coupler of the wheel trim cover includes a distal end, and wherein the distal end is spaced from the wheel when the wheel trim cover is attached to the wheel.

[0025] According to this embodiment, the wheel mold assembly includes: [0026] a first portion having a profile defining a front facing disc portion of the wheel; a second portion having a profile defining a rear facing disc portion of the wheel; and a plurality of side portions having a profile defining a rim portion of the wheel.

[0027] Various aspects of this invention will become apparent to those skilled in the art from the following detailed description of the preferred embodiments, when read in light of the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. **1** is a front elevational view of a cast vehicle wheel having a plurality of trim covers attached thereto and produced in accordance with the present invention.

[0029] FIG. **2** is a front elevational view of the wheel of FIG. **1** showing the trim covers of the wheel removed.

[0030] FIG. **3** is a perspective sectional view of the wheel taken along lines **3-3** in FIG. **1** illustrating a trim cover separated from the wheel.

[0031] FIG. **4** is a sectional view of the wheel taken along lines **4-4** in FIG. **2** illustrating a trim cover separated from the wheel.

[0032] FIG. **5** is an enlarged sectional view of a portion of the wheel of FIG. **4** with the trim cover attached to the wheel.

[0033] FIG. **6** is a partial cross-sectional view of a mold assembly having a tool insert illustrating the casting process for forming the wheel.

[0034] FIG. **7** is an enlarged top front perspective view of a tool insert used in the mold assembly of FIG. **6**.

[0035] FIG. **8** is bottom perspective view of the tool insert of FIG. **7**.

[0036] FIG. **9** is a partial cross-sectional view of the mold assembly and tool insert taken along lines **9-9** in FIG. **6**.

[0037] FIG. **10** is an enlarged top front perspective view of an alternate embodiment of a tool insert.

[0038] FIG. **11** is a schematic top view of an alternate embodiment of a tool insert.

[0039] FIG. **12** is a schematic top view of an alternate embodiment of a tool insert.

[0040] FIG. **13** is an enlarged top perspective view of an alternate embodiment of a tool insert.

[0041] FIG. **14** is an enlarged top perspective view of an alternate embodiment of a tool insert.

[0042] FIG. **15** is a perspective sectional view of an alternate embodiment of a wheel having a trim cover including three separate coupler mounting features.

[0043] FIG. **16** is a schematic front elevational view of a wheel.

[0044] FIG. **17** is a schematic front elevational view of a single piece trim cover.

[0045] FIG. **18** is a schematical front elevational view of the wheel of FIG. **16** having the single piece trim cover of FIG. **17** installed thereon.

[0046] FIG. **19** is a schematic front elevational view of another embodiment of a single piece trim cover.

[0047] FIG. **20** is a schematic front elevational view of the wheel of FIG. **16** having the single piece trim cover of FIG. **19** installed thereon.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0048] Referring now to the drawings, there is illustrated in FIG. **1** a vehicle wheel, indicated generally at **10**. The wheel **10** generally defines a wheel disc, indicated generally at **12**, and an outer annular rim, indicated generally at **14** (see FIGS. **3** and **4**). Although the invention is illustrated and described in conjunction with the particular vehicle wheel construction disclosed herein, it will be appreciated that the invention can be used in conjunction with other types of vehicle wheel constructions.

[0049] In a preferred embodiment (and as illustrated herein), the wheel disc **12** and the rim **14** are unitarily or monolithically produced, such as for example, as a single casting. After production of the casting, portions of the casting can be machined or otherwise worked to form the wheel **10**. Alternatively, the wheel disc **12** and rim **14** may be produced separately, such as separate castings and/or non-castings, and then joined together by any suitable means, to produce a “fabricated” vehicle wheel **10**. In this fabricated example, the wheel disc **12** is preferably produced as a single

casting.
[0050] The wheel **10** can be produced or cast from any suitable material. For example, the wheel **10** may be in all-cast wheel design formed from aluminum or alloys thereof. Aluminum is advantageous in that it is relatively inexpensive, lightweight, easily machinable, and can provide sufficient rigidity. Other suitable materials in which the wheel **10** can be made of include magnesium, titanium or alloys thereof, steel, carbon fiber and/or composite materials.
[0051] As shown in FIGS. **3** and **4**, the combination of the wheel disc **12** and the outer rim **14** defines a wheel axis X for the wheel **10**. The outer rim **14** can have any suitable shape for receiving and supporting a tire (not shown). In one embodiment, the outer rim **14** is machined after the wheel **10** is cast to provide a continuous annular shape relative to the wheel axis X for accommodating the tire. It should be appreciated that the outer rim **14** can have any desired diameter and/or shape.
[0052] The wheel disc **12** is generally comprised of a central hub **20** and a plurality of spokes **22** extending radially outwardly from the hub **20**. As stated previously, it is preferred that the wheel disc **12** is a single cast component such that the hub **20** and the spokes **22** are all formed together as a single cast.
[0053] The hub **20** is generally defined as a plate or disc defining a front face **26** (or front surface), as seen in FIGS. **1**, **2**, and **4**, and a rear face **28** (or rear surface), as seen in FIGS. **3** and **4**. The front face **26** is located on the outboard side of the wheel **10** when mounted on a vehicle. The rear face **28** is located on the inboard side of the wheel **10** when mounted on a vehicle. The hub **20** functions as a wheel mounting portion or center mounting portion of the wheel **10** for connecting with an axle (not shown) via a plurality of lug bolts (not shown) and lug nuts (not shown).
[0054] The hub **20** includes a centrally located pilot aperture or hub hole **30**. The hub hole **30** generally extends along the wheel axis X. The hub hole **30** may accommodate a portion of the axle and/or receive a protective/decorative cap (not shown). A plurality of lug bolt receiving holes **32** are circumferentially spaced around the hub hole **30** and wheel axis X. In the illustrated embodiment, the hub **20** includes five lug bolt receiving holes **32**. Alternatively, the number and/or location of the lug bolt receiving holes **32** may be other than illustrated if so desired. The lug bolt receiving holes **32** receive the lug bolts for securing the vehicle wheel **10** with lug nuts on the axle of an associated vehicle.
[0055] In the illustrated embodiment, the wheel disc **12** includes five relatively wide spokes **22** extending from the hub **20**. Alternatively, the number of the spokes **22** may be other than illustrated if so desired. For example, the vehicle wheel **10** may include less than five spokes **22** or more than five spokes **22**. The spokes **22** may have any suitable shape and/or decorative features formed therein, as shown in FIGS. **1** and **2**.
[0056] The wheel disc **12** further defines a plurality of vent holes **40** generally located between the spokes **22**. The vent holes **40** may have any suitable shape, such as a generally V-shape as shown in the figures. In the illustrated embodiment, the wheel disc **12** includes five vent holes **40** generally corresponding to the five spokes **22**. The wheel disc **12** further includes a plurality of flanges **42** generally located in the region of the vent holes **40** and extend from distal ends of the spokes **22**. In the illustrated embodiment, the wheel disc **12** includes ten flanges **42** in total such that a pair of flanges **42** are arranged in each region of the vent holes **40**. As will be discussed in detail below, each of the flanges **42** includes a mating feature formed therein for the attachment and positioning of a plurality of wheel trim covers **50**. It should be understood that the flanges **42** are not absolutely required and that the wheel **10** could be designed without the flanges **42** such that the mating features could be directly formed alternatively in other parts of the wheel **10**, such as in the spokes **22** or the hub **20**.
[0057] The wheel disc **12** includes the plurality of trim covers **50** that are attached to the wheel disc **12** after casting of the wheel **10**. In a preferred embodiment, the trim covers **50** are attached to the wheel disc **12** at radially outward portions of the vent holes **40**, as best shown in FIG. **1**. Note that the wheel **10** illustrated in FIG. **2** does not show any trim covers **50** attached thereto. Of course, the

trim covers **50** can have any suitable shape and may be mounted in other suitable locations on the wheel disc **12** other than as illustrated. In the illustrated embodiment, the wheel disc **12** includes five trim covers **50** generally corresponding to the number of vent holes **40** and spokes **22**. Instead of having a plurality of trim covers **50**, it should be understood that a single trim cover may be utilized, as will be explained below with respect to FIGS. **16** through **20**.

[0058] In a preferred embodiment, the trim covers **50** are made of a generally lightweight plastic material. The trim covers **50** may be simply added to the wheel **10** to provide an aesthetically pleasing appearance. Alternatively, or in addition to, the trim covers **50** may provide for an aerodynamic enhanced characteristic of the wheel **10**. The trim covers **50** may provide for a more “bulkier” appearance without adding much additional weight to the wheel **10** compared to providing casted metal material in the same region of the wheel disc **12**. The trim covers **50** may have a color different from the color of the wheel **10**, or may have a similar color as to blend in aesthetically.

[0059] As stated above, the trim covers **50** may have any suitable shape. In the illustrated embodiment, the trim covers **50** have a somewhat flattened slightly arcuate shape that extends between distal ends of adjacent spokes **22**. As best shown in FIG. **3**, the trim covers **50** can have any suitably shaped ridges to provide rigidity to the trim covers **50**. Each of the trim covers **50** is attached to the wheel disc **12** by two threaded fasteners or bolts **54**. The bolts **54** may be a bolt, a fastener or a self-tapping screw, for example. Of course, the trim covers **50** may be attached to the wheel disc **12** by a single bolt **54** or more than two bolts (see the embodiment illustrated in FIG. **15**). Alternatively, the trim covers **50** may be attached to the wheel disc **12** by fasteners (not shown) other than the bolts **54**, such as by rivets, clips, or adhesive. As discussed below, during attachment of the trim covers **50**, the bolts **54** are received in bores or pass through bolt holes **56** formed in the flanges **42**. The bolt holes **56** may be formed from a machining or drilling operation after the wheel **10** has been cast. Alternatively, the bolt holes **56** may be formed from integral features of a mold assembly used to cast the wheel **10**

[0060] Referring to FIG. **5**, a more detailed illustration of a portion of a trim cover **50** is provided. As stated above, each trim cover **50** is attached to the wheel disc **12** at two locations, indicated generally and defined as mating features, indicated generally at **60**. One such location is shown in the sectional drawing of FIG. **5**. The mating feature **60**, as shown in the illustrated embodiment, is in the form a coupler **62**. The coupler **62** may be provided as a bulge, mound or protuberance extending rearwardly (upwardly as viewing FIG. **5**) from the trim cover **50**. The coupler **62** may have a cylindrical shape defining a diameter D. Of course, the coupler **62** may have any suitable shape. In the illustrated embodiment, the coupler **62** includes a rounded end **64**. Of course, the end **64** need not be rounded and/or may include a chamfer instead.

[0061] As will be discussed in detail below, the couplers **62** of the trim covers **50** provide means for centering or aligning the trim cover **50** relative to the wheel disc **12** when the trim covers **50** are attached to the wheel disc **12**. The couplers **62** are received in corresponding mating features, indicated generally at **70**, of the wheel disc **12**. More specifically, in the illustrated embodiment shown in FIG. **5**, the mating features **70** of the wheel disc **12** are in the form of holes or recesses **72** formed in the flanges **42**. The recesses **72** may be through holes or non-through hole bores formed in the flanges **42**. The recesses **72** may have a cylindrical shape generally corresponding to the shape and profile of the couplers **62**. In the illustrated embodiment, the recesses **72** have a diameter DD. The diameter DD of the recesses **72** may be relatively the same size of the diameter D of the corresponding couplers **62**. The diameter D may be slightly greater than the diameter DD such that a frictional fit is created when a coupler **62** is inserted into a corresponding recess **72**. Alternatively, the diameter D may be slightly smaller than the diameter DD such that a slight cylindrical gap may exist between the walls of the couplers **62** and the recesses **72**, such as shown in FIG. **5**. Note that this gap is shown highly exaggerated in the drawing of FIG. **5** for clarity purposes and is not representative of the actual gap size. As is also shown in FIG. **5**, the gap between the walls of the couplers **62** and the recesses **72** may be slightly non-symmetrical such that the couplers **62** are not completely centered within the corresponding recesses **72**. This misalignment may be caused when the bolt **54** is threaded into a threaded bore **76** formed in the corresponding coupler **62** during attachment of the trim cover **50** to the wheel disc **12**.

[0062] To attach a trim cover **50** to a corresponding portion of the wheel disc **12**, the trim cover **50** is positioned such that coupler **62** is inserted into the corresponding recess **72**. Note that for each of the trim covers **50**, two couplers **62** are placed with a corresponding pair of recesses **72**. Note that a distal end **78** of the coupler **62** may contact a bottom surface **80** of a corresponding recess **72**, as shown in FIG. **5**. Alternatively, a small gap may exist between the distal end **78** and the bottom surface **80** such that the distal end **78** is spaced from the bottom surface **80**. Also note that other various structures **82** of the trim cover **50** may contact and engage with portions of the wheel disc **12**. Alternatively, the various structures **82** may be designed such that there is no contact to reduce rattling, for example. Once the coupler **62** is positioned within the corresponding recess **72**, one of the bolts **54** may be inserted through the corresponding bolt hole **56** of the flange **42** and then threaded into the threaded bore **76** of the coupler **62**, thereby fastening the trim cover **50** to the wheel disc **12**. Alternatively, self-tapping screws (not shown) may be utilized. For example, there could be a lack of a threaded bore **76** and the bolts **54** may be replaced with self-tapping screws or bolts which are inserted into the trim cover **50**. In this case, the thread may be made when the self-tapping screw is used for the first time.

[0063] As will be discussed below, slight manufacturing tolerances in the casting wheel **10** and/or the formation of the trim cover **50** may provide for a slight misalignment of the trim cover **50** from a desired first predetermined position relative to the wheel disc **12**. As will be discussed in detail below, a method of manufacturing the wheel **10** in accordance with the present invention will assist in adjusting/correcting this misalignment in a low-cost manner.

[0064] Referring now to FIG. **6**, a preferred method of manufacturing the wheel **10** will now be described. FIG. **6** illustrates a portion of a mold assembly, indicated generally at **90**, for use in casting the wheel **10**. The section of FIG. **6** is generally taken through the portion of the cast wheel **10** around one of the flanges **42**. The wheel **10** may be cast in any suitable manner and it should be understood that the method described herein with respect to the mold assembly **90** is just one suitable method for producing the wheel **10**. The mold assembly **90** includes a first portion **92** having a surface **94** with a profile that generally corresponds to the front face **26** of the wheel disc **12** of the wheel **10**. The mold assembly **90** further includes a second portion **96** having a surface **98** with a profile that generally corresponds to the rear face **28** of the wheel disc **12** of the wheel **10**. The surface **98** may also extend (upward as viewing FIG. **6**) such that it corresponds with an inner portion of the rim **14**. The mold assembly **90** further includes one or more side portions **100** having surfaces **102** with a profile that generally corresponds to the outer portion of the rim **14**. When the mold assembly **90** is in a closed position, as shown in FIG. **6**, the first, second, and side portions **92**, **96**, and **100** form a cavity **104** generally defining the shape of the wheel **10**. Note that in FIG. **6**, the cavity **104** is shown filled with a cast material. It is also noted that a plurality of broken or phantom lines **106** represent the final shape of the wheel **10** after a machining process is subsequently performed to the casting of the wheel **10** to provide the desired and smoothed shape of the wheel **10**.

[0065] In accordance with a preferred embodiment of the present invention, the mold assembly **90** includes a plurality of removable tool inserts, indicated generally at **110**. The inserts **110** are positioned within the mold assembly **90** and utilized to form the recesses **72** in the flanges **42** of the wheel disc **12**. As will be discussed further below, differently shaped removable inserts **110** (such as shown in FIG. **10**) can be replaced and exchanged within the mold assembly **90** to alter the formation of the corresponding recesses **72** to change the desired size, shape and location of the recesses **72**. Note that only one insert **110** is shown in the sectional drawing of FIG. **6**. Preferably, ten inserts **110** are used within the mold assembly **90** corresponding to the ten recesses **72** of the wheel disc **12**. The inserts **110** are disposed within bores **112** formed in the first portion **92** of the mold assembly **90**. The bores **112** may be multi-stepped bores for accommodating the inserts **110** as well as the mounting bolts **114**. The mounting bolts **114** secure the inserts **110** within the first portion **92** of the mold assembly **90**. Of course, other means for securing the inserts **110** may be used other than the mounting bolts **114**.

[0066] Referring now to FIGS. **7** and **8**, an initial or first tool insert **110** is shown in greater detail. The insert **110** includes a cylindrically shaped main body **120**. The body **120** defines a proximal end **122** and a distal end **124**. A threaded bore **126** is formed in the proximal end **122** of the insert **110**. The threaded bore **126** receives a corresponding mounting bolt **114** to secure the insert **110** within the first portion **92** of the mold assembly **90**. The threaded bore **126** may be axially centered within the cylindrical main body **120**.

[0067] In a preferred embodiment, the insert **110** does not rotate within the corresponding bore **112** formed in the first portion **92** of the mold assembly once the insert **110** is mounted therein. This non-rotational mounting may be provided by any suitable manner. For example, as shown in FIGS. **7** and **8**, the insert **110** may be provided with a keyway or notch **128** formed in the proximal end **122** thereof. The keyway **128** interacts with a ledge **130** (see FIG. **6**) radially inwardly formed within the bore **112**, thereby preventing rotation of the insert **110** when the insert **110** is installed into the bore **112**. Although the notch **128** is illustrated with three surfaces, it should be understood that it may be made more simply with a single surface, or alternatively may have any suitable shape that interacts with the ledge **130** to prevent rotation.

[0068] Located at the distal end **124** of the insert **110** is a mounting feature provided in the form or a dome or protrusion **140**. The protrusion **140** communicates with and extends into the cavity **104** of the mold assembly **90** when the insert **110** is mounted within the bore **112**, as shown in FIG. **6**. The shape and size of the protrusion **140** corresponds to the shape and size of the recess **72** formed in the casted wheel disc **12**. Thus, the protrusion **140** creates the recess **72** when the wheel **10** is cast within the mold assembly **90**. Of course, the physical attributes of the protrusion **140** and the recess **72** may be switched such that the insert **110** has a recess which receives a protrusion formed in the wheel disc **12**.

[0069] The protrusion **140** can have any suitable shape. For example, in the embodiment shown in FIGS. **6** through **8**, the protrusion **140** has a cylindrical shape corresponding to the shape of the recess **72**. The shape of the recess **72** corresponds to shape of the coupler **62** of the trim cover **50** when the trim cover **50** is mounted on the wheel disc **12**. The protrusions **140**, the recesses **72**, and the couplers **62** can have any corresponding shape and need not be cylindrical. For example, they may each have a frustoconical shape. In another example, there is illustrated in FIG. **13** and alternate embodiment of an insert **150** including a protrusion **152** having a rectangular block shape. Accordingly, the insert **150** would be used with a trim cover **50** having a coupler with a corresponding rectangular block shape. There is illustrated in FIG. **14** of yet another alternate embodiment of an insert **154** including a protrusion **156** having a triangular or pyramidal shape.

[0070] The preferred method of manufacturing the wheel **10** will now be discussed. A plurality of trim covers **50** is provided having a desired shape such as the shape of the trim covers **50** illustrated in the figures. In general, the trim covers **50** have been manufactured, such as by a suitable plastic forming process, and will generally not be replaced or remanufactured to a different specification. However, it should be understood that due to changing manufacturing tolerances in the production of the trim covers **50**, the final size and shape of the trim covers **50** may vary slightly from the initially designed size and shape.

[0071] A first set of initially designed tool inserts **110** are installed into the wheel mold assembly **90**. More precisely, ten first tool inserts **110** are installed into the corresponding ten bores **112** of the first portion **92** of mold assembly **90**. These initially designed first tool inserts **110** may have the general shape as the inserts **110** illustrated herein. Once installed, the protrusions **140** of the inserts **110** will be introduced within the cavity **104** of the mold assembly **90**. The mold assembly **90** is closed and casting material is introduced into the cavity **104** of the mold assembly **90**, such as is shown in FIG. **6**. During the casting process, recesses **72** will be formed in accordance with the shape, size and profile of the protrusions **140** of the inserts **110**.

[0072] The wheel **10** is then removed from the mold assembly **90**. The trim covers **50** are then optionally installed onto the wheel **10** such that the couplers **62** of the trim covers **50** are inserted into corresponding formed recesses **72** of the wheel disc **12**. The trim covers **50** should be properly positioned relative to the wheel disc **12** in a first predetermined position per the initially designed configuration. The position of the trim covers **50** is then evaluated and measured to determine if the trim covers **50** are, in fact, within the desired first predetermined designed position. If it is determined that the trim covers **50** are not properly positioned within acceptable specifications, the initially formed wheel **10** may be discarded. Secondary tool inserts **110′**, such as is shown schematically in FIG. **10**, are then provided to replace the initially designed first tool inserts **110** for correcting any misalignment that may have occurred. The secondary inserts **110′** are designed such that newly designed protrusions **140′** are provided for producing differently positioned and/or sized recesses **72′** in a secondary formed wheel **10′**. It is noted that the differently shaped protrusion **140′** of the secondary insert **110** illustrated in FIG. **10** is highly exaggerated for clarity purposes and that in reality, only minor changes will generally be required. As shown schematically in FIG. **10**, the protrusion **140′** may be designed larger or smaller and/or offset relative to the cylindrical main body **120′**. FIG. **11** is a schematic illustration of a top view of the first initially designed insert **110** showing the protrusion **140** being generally axially aligned therewith. FIG. **12** is a highly exaggerated schematic illustration of the secondary designed inserts **110′** having a protrusion **140′** designed smaller and offset to manage the minor misalignment issues of the installed placement of the trim covers **50** relative to the wheel disc **10**.

[0073] The first tool inserts **110** are exchanged with the newly designed and formed secondary inserts **110′** within the first portion **92** of the mold assembly **90**. The protrusions **140′** of the new inserts **140′** will be designed to take into consideration the misalignment determined in the evaluation and measuring step as described above. A second wheel is cast within the mold assembly **90** and trim covers **50** are installed thereon. The wheel **10** may then be reevaluated and measured to confirm that the proper placement of the trim covers **50** relative to the wheel are accurate and are within a second predetermined position. Thus, the use of the secondary exchanged tool inserts **110′** assist to improve the centering and positioning of the trim covers **50**, thereby avoiding a costly procedure of re-machining, retooling, or replacing portions of the mold assembly **90**. The use of the secondary exchanged tool inserts **110′** described above may also avoid costly changeover expenses. The wheels **10** may have variations caused by differences in material shrinkage behavior or by mold wear, and thus, the secondary exchanged tool inserts **110′** could be also be utilized for mold wear reasons.

[0074] It should be understood that installing the trim covers **50** onto the wheel **10** prior to evaluating the positional dimensions and determining the positional accuracy of the mounting feature of the first tool inserts **100** is not required and is an optional step. The evaluation may be determined even if the trim covers **50** are not installed onto the wheel **10**. Evaluation of the positional dimensions may be done with the assistance of a coordinate measuring machine (CMM) (not shown), which is well known in the industry. A CMM is typically a device that measures the geometry of physical objects by sensing discrete points on the surface of the object with a probe. The machine generally specifies the position of the probe in terms of displacement from an origin point in a three-dimensional coordinate system (XYZ axes). Various types of probes may be used with the CMM, such as for example, mechanical, laser, other optical and white light sensors.

[0075] As stated above, the trim covers **50** may have any suitable shape and may be provided with any number of fasteners for attachment to the wheel **10**. There is illustrated in FIG. **15** an alternate embodiment of a wheel **10″** illustrating that a differently shaped trim cover **50″** uses three separate bolts **54″** for attachment to the wheel **10″**.

[0076] As also stated above, instead of having a plurality of trim covers **50**, a single trim cover may be used to cover desired locations of a wheel face. FIGS. **16** through **20** schematically demonstrate the use of a single trim cover per wheel. There is schematically illustrated in FIG. **16** a wheel, indicated generally at **200**. The wheel **200** may have any suitable design. In the schematic illustrated embodiment, the wheel **200** includes a plurality of spokes **202** and vent holes **204**. The illustrated wheel **200** has a five spoke design including openings **206** formed the spokes **202**. Of course, the spokes **202** have any suitable shape and may be formed with the openings **206**.
[0077] There is schematically illustrated in FIG. **17** a single trim cover, indicated generally at **210**, for mating with and installed onto the wheel **200** by the method described above. Thus, the wheel **200** and the trim cover **210** may have mating features (not shown) and couplers (not shown) as described above for attaching the trim cover **210** to the wheel **200**. The method of manufacturing using inserts, mold assemblies, and evaluating techniques describe above may be utilized in manufacturing the wheel **200** and trim cover **210**. The trim cover **210** includes spoke portions **212** (five shown) extending radially outwardly from a hub **214**. Note that outer ends **216** of the spoke portions **212** are not connected together. Of course, the trim cover **210** could be designed such that the outer ends **216** (or any other portions of the trim cover **210**) are connected together. FIG. **18** schematically illustrates the trim cover **210** mounted on the wheel **200**. The spoke portions **212** of the trim cover **210** generally cover the openings **206** of the spokes **202** of the wheel **200**. Of course, the wheel **200** and the trim cover **210** may have suitable design and the spoke portions **212** of the trim cover **210** need not cover the openings **206** of the spokes **202** of the wheel **200**.
[0078] There is schematically illustrated in FIG. **19** another embodiment of a single trim cover, indicated generally at **230**, for mating with and installed onto the wheel **200**. Similar to the trim cover **210**, the trim cover **230** may have mating features (not shown) and couplers (not shown) as described above for attaching the trim cover **210** to the wheel **200**. The method of manufacturing using inserts, mold assemblies, and evaluating techniques describe above may be utilized in manufacturing the wheel **200** and trim cover **230**. The trim cover **230** includes spoke portions **232** (five shown) extending radially outwardly from a hub **234**. The spoke portions **232** have outer ends **236** which are connected to an outer circumferential band **238**. The circumferential band **238** can have any suitable width or shape. FIG. **20** schematically illustrates the trim cover **230** mounted on the wheel **200**. Similar to the trim cover **210**, the trim cover **230** generally covers the openings **206** of the spokes **202** of the wheel **200**.
[0079] The principle and mode of operation of this invention have been explained and illustrated in its preferred embodiments. However, it must be understood that this invention may be practiced otherwise than as specifically explained and illustrated without departing from its spirit or scope.

Claims

1. A method of manufacturing a cast vehicle wheel including at least one wheel trim cover attached thereto, wherein the method comprises the steps of: (a) providing a wheel mold assembly having a cavity defining a shape of the wheel; (b) providing a removable first tool insert adapted to be installed into the mold assembly, wherein the first tool insert includes a mounting feature that communicates with the cavity of the mold assembly to form a mating feature in the wheel when the wheel is cast; (c) installing the first tool insert into the mold cavity; (d) casting the wheel by introducing a casting material into the cavity of the mold assembly, thereby forming the mating feature in the wheel; (e) removing the wheel from the mold assembly; (f) providing at least one wheel trim cover adapted to be attached to the wheel at the mating feature of the wheel via a coupler formed in the wheel trim cover such that a relative position of the coupler and the mating feature positions the wheel trim cover relative to the wheel in a first predetermined position; (g) optionally attaching the wheel trim cover onto the wheel at a location of the coupler and the mating feature; (h) evaluating positional dimensions of the mating feature of the wheel relative to the

coupler of the wheel trim cover to determine a positional accuracy of the mounting feature of the first tool insert; (i) subsequent to step (h), providing a second tool insert having a second mounting feature dimensionally different from the mounting feature of the first tool insert which provides improved positional accuracy of the wheel trim cover relative to the wheel in a second predetermined position; (j) replacing the first tool insert from the mold cavity and installing the second tool insert into the mold assembly; (k) casting a second wheel by introducing a casting material into the cavity of the mold assembly; and (l) attaching the wheel trim cover onto the second wheel at the location of the coupler and the second mating feature.
2. The method of claim 1, wherein the coupler of the wheel trim cover is in the form of a protrusion, and wherein the mating feature of the wheel is a recess receiving the protrusion when the wheel trim cover is attached to the wheel.
3. The method of claim 2, wherein in step (d) during the casting process, the formation of the profile of the mating feature of the wheel corresponds to the profile of the mounting feature of the first tool insert.
4. The method of claim 1, wherein in step (h) a coordinate measuring machine is used to evaluate positional dimensions of the mating feature of the wheel relative to the coupler of the wheel trim cover to determine a positional accuracy of the mounting feature of the first tool insert.
5. The method of claim 1, wherein in step (l) the wheel trim cover is attached onto the second wheel by at least one fastener, and wherein the fastener is a threaded bolt.
6. The method of claim 5, wherein prior to step (l) a hole is formed in the second wheel at the second mating feature of the second wheel for receiving the threaded bolt.
7. The method of claim 6, wherein the hole is formed by a machining or drilling process after the wheel is cast.
8. The method of claim 6, wherein the hole is formed in the casting process.
9. The method of claim 6, wherein the wheel trim cover is provided with internal threads formed therein for receiving the threaded bolt.
10. The method of claim 9, wherein the coupler of the wheel trim cover is provided with the internal threads.
11. The method of claim 10, wherein the coupler is a protrusion formed in the wheel trim cover.
12. The method of claim 11, wherein the second mating feature of the second wheel is a recess receiving the protrusion of the wheel trim cover.
13. The method of claim 12, wherein the protrusion is cylindrically shaped.
14. The method of claim 13, wherein the recess is cylindrically shaped.
15. The method of claim 1, wherein in step (l) the wheel trim cover is attached onto the second wheel by at least one fastener.
16. The method of claim 1, wherein in step (h), evaluating the positional dimensions of the mating feature of the wheel relative to the coupler of the wheel trim cover includes measuring the position of the mating feature of the wheel relative to the position of the coupler of the wheel trim cover.
17. The method of claim 1, wherein the tool insert includes a keyway which mates with a key portion of the mold assembly to prevent rotation of the tool insert within the mold assembly.
18. The method of claim 1, wherein the coupler of the wheel trim cover includes a distal surface, and wherein the distal surface contacts the wheel when the wheel trim cover is attached to the wheel.
19. The method of claim 1, wherein the coupler of the wheel trim cover includes a distal end, and wherein the distal end is spaced from the wheel when the wheel trim cover is attached to the wheel.
20. The method of claim 1, wherein the wheel mold assembly includes: a first portion having a profile defining a front facing disc portion of the wheel; a second portion having a profile defining a rear facing disc portion of the wheel; and a plurality of side portions having a profile defining a rim portion of the wheel.
21. A cast vehicle wheel produced according to the steps of claim 1.